

Qualification Pack



Automotive Robotics System Integrator/Planner

QP Code: ASC/Q8304

Version: 1.0

NSQF Level: 6

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Qualification Pack

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ASC/Q8304: Automotive Robotics System Integrator/Planner

Brief Job Description

The individual is primarily involved in installation, interfacing and programming processes of industrial robot and cobot systems. They support the robot technician in activities such as robotic cell anatomy, robotic cell layout mapping and development, wire harnessing, interfacing and installation of robot/cobot setups and their programming.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9810: Manage work and resources \(Manufacturing\)](#)
2. [ASC/N9812: Interact effectively with team, customers and others](#)
3. [ASC/N8316: Identify product feasibility and setup requirements](#)
4. [ASC/N8317: Selection and setup of end-effector and robot](#)
5. [ASC/N8318: Installation, commissioning and integration of robot system](#)
6. [ASC/N8319: Robot/Cobot programming and application testing](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Research & Development
Occupation	Automotive Product Development
Country	India
NSQF Level	6
Credits	19

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/1223.0102
Minimum Educational Qualification & Experience	Certificate-NSQF (Automotive Prototype Manufacturing Lead Technician Level 5) with 3 Years of experience of relevant experience OR M.E. (in the relevant field) with 1 Year of experience in the relevant field OR M.Tech (in the relevant field) with 1 Year of experience in the relevant field OR B.E./B.Tech (in the relevant field) with 2 Years of experience in the relevant field OR Diploma (3 years in Mechanical/Automobile/ Electrical / Electronics from recognized regulatory body after class 12th) with 4 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	28/07/2025
NSQC Approval Date	28/07/2022
Version	1.0
Reference code on NQR	2022/AUT/ASDC/06128
NQR Version	1.1

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ASC/N9810: Manage work and resources (Manufacturing)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain Health and Hygiene
- Effective waste management practices
- Material/energy conservation practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices for dealing with hazards to ensure safety of self and others
- PC3.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC4.** ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person

Maintain Health and Hygiene

To be competent, the user/individual on the job must be able to:

- PC9.** ensure workplace, equipment, restrooms etc. are sanitized regularly
- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC12.** report advanced hygiene and sanitation issues to appropriate authority
- PC13.** follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc
- PC14.** wear and dispose PPEs regularly and appropriately

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Effective waste management practices

To be competent, the user/individual on the job must be able to:

PC15. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

PC16. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

PC17. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

PC18. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

PC19. identify possibilities of using renewable energy and environment friendly fuels

PC20. identify processes where material and energy/electricity utilization can be optimized

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, individual role and responsibilities in this context

KU2. the organisation's emergency procedures for different emergency situations and the importance of following the same

KU3. evacuation procedures for workers and visitors

KU4. how and when to report hazards as well as the limits of responsibility for dealing with hazards

KU5. potential hazards, risks and threats based on the nature of work

KU6. various types of fire extinguisher

KU7. various types of safety signs and their meaning

KU8. appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

KU9. relevant standards, procedures and policies related to 5S followed in the company

KU10. the various materials used and their storage norms

KU11. importance of efficient utilisation of material and water

KU12. basics of electricity and prevalent energy efficient devices

KU13. common practices of conserving electricity

KU14. common sources and ways to minimize pollution

KU15. categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

KU16. waste management techniques

KU17. significance of greening

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	20	13	-	8
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	4	2	-	2
PC2. implement safe working practices for dealing with hazards to ensure safety of self and others	3	1	-	2
PC3. conduct regular checks of the machines with support of the maintenance team to identify potential hazards	2	2	-	1
PC4. ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions	3	2	-	1
PC5. organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices	2	-	-	-
PC6. fill daily check sheet to report improvements done and risks identified	2	2	-	-
PC7. ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others	2	2	-	1
PC8. report any identified breaches in health, safety and security policies and procedures to the designated person	2	2	-	1
<i>Maintain Health and Hygiene</i>	13	7	-	5
PC9. ensure workplace, equipment, restrooms etc. are sanitized regularly	3	2	-	1
PC10. ensure team is aware about hygiene and sanitation regulations and following them on the shop floor	2	1	-	-
PC11. ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. report advanced hygiene and sanitation issues to appropriate authority	1	1	-	1
PC13. follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc	2	1	-	1
PC14. wear and dispose PPEs regularly and appropriately	3	-	-	1
<i>Effective waste management practices</i>	6	4	-	1
PC15. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP	3	2	-	-
PC16. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste	3	2	-	1
<i>Material/energy conservation practices</i>	11	6	-	6
PC17. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively	2	2	-	1
PC18. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water	3	2	-	1
PC19. identify possibilities of using renewable energy and environment friendly fuels	3	1	-	2
PC20. identify processes where material and energy/electricity utilization can be optimized	3	1	-	2
NOS Total	50	30	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9810
NOS Name	Manage work and resources (Manufacturing)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Qualification Pack

ASC/N9812: Interact effectively with team, customers and others

Description

This unit is about communicating with team members, superior and others.

Scope

The scope covers the following :

- Communicate effectively with team members
- Interact with superiors
- Respect gender and ability differences

Elements and Performance Criteria

Communicate effectively with team members

To be competent, the user/individual on the job must be able to:

- PC1.** implement ways to share information with team members in line with organisational requirements
- PC2.** ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written
- PC3.** manage and co-ordinate with team members to integrate work as per requirements
- PC4.** work in a way that show respect for all team members and customers
- PC5.** carry out commitments made to team members and let them know in good time if there is any discrepancy with reasons
- PC6.** resolve conflicts within the team members at work to achieve smooth workflow
- PC7.** guide the team members to follow the organisation's policies and procedures
- PC8.** ensure team goals are given preference over individual goals
- PC9.** respect personal space of colleagues and customers

Interact with superiors

To be competent, the user/individual on the job must be able to:

- PC10.** report progress on job allocated and team performance to the superiors
- PC11.** escalate problems to superiors that cannot be handled
- PC12.** train the team members to report completed work and receive feedback on work done
- PC13.** encourage team members to rectify errors as per feedback and minimize mistakes in future

Respect gender and ability differences

To be competent, the user/individual on the job must be able to:

- PC14.** ensure team shows sensitivity towards all genders and PwD
- PC15.** adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability
- PC16.** help PwD team members to overcome the challenges, if asked

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with team members and superiors
- KU2.** different methods of communication as per the circumstances
- KU3.** gender based concepts, issues and legislation
- KU4.** organisation standards and guidelines to be followed for PwD
- KU5.** rights and duties at workplace with respect to PwD
- KU6.** organisation policies and procedures pertaining to written and verbal communication

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** read instructions/guidelines/procedures
- GS7.** write in English/any one language

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with team members</i>	20	14	-	8
PC1. implement ways to share information with team members in line with organisational requirements	2	2	-	-
PC2. ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written	2	2	-	2
PC3. manage and co-ordinate with team members to integrate work as per requirements	2	1	-	2
PC4. work in a way that show respect for all team members and customers	3	1	-	2
PC5. carry out commitments made to team members and let them know in good time if there is any discrepancy with reasons	2	2	-	-
PC6. resolve conflicts within the team members at work to achieve smooth workflow	3	2	-	-
PC7. guide the team members to follow the organisation's policies and procedures	2	1	-	-
PC8. ensure team goals are given preference over individual goals	2	1	-	-
PC9. respect personal space of colleagues and customers	2	2	-	2
<i>Interact with superiors</i>	18	10	-	7
PC10. report progress on job allocated and team performance to the superiors	4	3	-	2
PC11. escalate problems to superiors that cannot be handled	4	2	-	1
PC12. train the team members to report completed work and receive feedback on work done	5	2	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. encourage team members to rectify errors as per feedback and minimize mistakes in future	5	3	-	2
<i>Respect gender and ability differences</i>	12	6	-	5
PC14. ensure team shows sensitivity towards all genders and PwD	4	2	-	2
PC15. adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability	4	2	-	2
PC16. help PwD team members to overcome the challenges, if asked	4	2	-	1
NOS Total	50	30	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9812
NOS Name	Interact effectively with team, customers and others
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

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ASC/N8316: Identify product feasibility and setup requirements

Description

This NOS unit is about developing the feasibility report and setup finalization for automobile manufacturing process, defining and calculating the numbers of equipment and resources needed to commission the process.

Scope

The scope covers the following :

- Feasibility study and report generation
- Application identification
- Setup and equipment identification
- Work load and payload calculations

Elements and Performance Criteria

Feasibility study and report generation

To be competent, the user/individual on the job must be able to:

- PC1.** read and interpret product documents like need analysis, feasibility, technical specification, process flow diagram, product drawings and other engineering documents to prepare/interpret the project design
- PC2.** identify all the components to be joined in a particular production cell and inputs and outputs in a robotic cell
- PC3.** identify assembly plan and sequence of operations to be perform for integrating the system
- PC4.** define material loading and unloading sequence in the robotic cell

Application identification

To be competent, the user/individual on the job must be able to:

- PC5.** collect and interpret the data of production volume and time available for the production
- PC6.** perform mathematical calculations on given data to calculate total work to be done to size the production line
- PC7.** interpret the application to be implemented on robotic system from the project documents
- PC8.** calculate other jigs/fixture and equipment required during the assembly process
- PC9.** define standard work cycle of the system

Setup and equipment identification

To be competent, the user/individual on the job must be able to:

- PC10.** read components' drawings and production drawings to identify the machine type and equipment to be used in the application
- PC11.** identify specification and quantity of material required for the production process
- PC12.** define process flow diagram of the production process
- PC13.** assess the designed process and identify potential failures in it by following organizational procedures

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- PC14.** perform payload calculation of the process for the selection of robot
- PC15.** interpret production volume and product size from the product design and project documents
- PC16.** identify process repeatability and cycle time

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** use of electronic equipment like computers and printers
- KU3.** data safety and non-disclosers norms
- KU4.** cyber safety and work confidentiality good practices
- KU5.** importance of different documents involved in product development
- KU6.** BIW Structure and different joining technologies
- KU7.** classification of the automation elements as power and safety elements (electrical incomer, circuit breakers, compressed air, hydraulic power pack, FRL, pressure relief valve etc.), input elements (proximity sensors, push buttons, limit switches, reed switches), control elements (relay, contactors, VFD, HMI, pneumatic and hydraulic solenoid valves) and output elements (indicators, buzzer, induction motors, pneumatic and hydraulic actuators)
- KU8.** types of control system used in the automation system
- KU9.** installation process includes mounting, wiring standards, routing, element assembly
- KU10.** programming of PLC and simulation tools from different makers along with integration of automation elements
- KU11.** calculation of cycle time of process
- KU12.** procedure of developing a manufacturing process
- KU13.** possible failures of automation system
- KU14.** importance of different documents involved in product development
- KU15.** robot work environment and conditions of operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations

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GS9. check the work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Feasibility study and report generation</i>	12	12	-	6
PC1. read and interpret product documents like need analysis, feasibility, technical specification, process flow diagram, product drawings and other engineering documents to prepare/interpret the project design	3	3	-	1
PC2. identify all the components to be joined in a particular production cell and inputs and outputs in a robotic cell	3	3	-	2
PC3. identify assembly plan and sequence of operations to be perform for integrating the system	3	3	-	2
PC4. define material loading and unloading sequence in the robotic cell	3	3	-	1
<i>Application identification</i>	12	12	-	6
PC5. collect and interpret the data of production volume and time available for the production	2	2	-	1
PC6. perform mathematical calculations on given data to calculate total work to be done to size the production line	2	2	-	2
PC7. interpret the application to be implemented on robotic system from the project documents	3	3	-	1
PC8. calculate other jigs/fixture and equipment required during the assembly process	3	3	-	1
PC9. define standard work cycle of the system	2	2	-	1
<i>Setup and equipment identification</i>	16	16	-	8
PC10. read components' drawings and production drawings to identify the machine type and equipment to be used in the application	2	2	-	1
PC11. identify specification and quantity of material required for the production process	3	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. define process flow diagram of the production process	3	3	-	1
PC13. assess the designed process and identify potential failures in it by following organizational procedures	2	2	-	1
PC14. perform payload calculation of the process for the selection of robot	2	2	-	1
PC15. interpret production volume and product size from the product design and project documents	2	2	-	1
PC16. identify process repeatability and cycle time	2	2	-	1
NOS Total	40	40	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8316
NOS Name	Identify product feasibility and setup requirements
Sector	Automotive
Sub-Sector	Research & Development
Occupation	Automotive Product Development
NSQF Level	6
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

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ASC/N8317: Selection and setup of end-effector and robot

Description

This NOS unit is about performing tasks related to selection and setup of robots and end-effector.

Scope

The scope covers the following :

- Robot and EOAT (End of arm tooling) selection
- Layout marking of robotic cell and tool assembly
- Positioning of equipment/components in the cell

Elements and Performance Criteria

Robot and EOAT Selection

To be competent, the user/individual on the job must be able to:

- PC1.** identify profile of the product panel and application of the robot in it by interpreting the process documents
- PC2.** read manual and technical specification of robots and define the requirements for the robot needed
- PC3.** select the robot on the basis of reachability requirements and accuracy requirements of the robot in the application
- PC4.** select the EOAT on the basis of its capability of handling maximum load
- PC5.** identify the zoning area and stroke area of robot by interpreting the process documents
- PC6.** determine the application controllers and external I/O devices required as per the project specifications and requirements

Layout marking of robotic cell and tool assembly

To be competent, the user/individual on the job must be able to:

- PC7.** determine area required for system implementation and availability of power, pneumatic and coolant supply
- PC8.** plan material space, trolleys, supply of material to line side & material handling equipment
- PC9.** find the position of equipment and finalize the robot positions according to it

Positioning of components in the cell

To be competent, the user/individual on the job must be able to:

- PC10.** identify the mounting and Tool Center Point (TCP) of equipment
- PC11.** finalize the required work tables of fixtures, orientation of loading and unloading and material flow in the robotic cell
- PC12.** mount and place all the components of robotic cell like robot, tip dressers, jigs/fixture/grippers, docking units, sensor and cable trays etc. as per the design document

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** software and 3D tools used in organisation
- KU3.** basics of electrical safety
- KU4.** safe operation of electronic equipment like computers and printers
- KU5.** data safety and non-disclosers norms
- KU6.** cyber safety and work confidentiality good practices
- KU7.** robot anatomy and Robot applications
- KU8.** EOAT anatomy and pneumatic systems
- KU9.** pay load requirements, reachability requirements and accuracy requirements
- KU10.** criteria and parameters for the selection of robot, EOAT and other accessories needed
- KU11.** procedure of designing and layouting of robotic cell and its positions
- KU12.** criteria for writing the new equipment specification manual

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Robot and EOAT Selection</i>	16	16	-	8
PC1. identify profile of the product panel and application of the robot in it by interpreting the process documents	3	3	-	2
PC2. read manual and technical specification of robots and define the requirements for the robot needed	3	2	-	1
PC3. select the robot on the basis of reachability requirements and accuracy requirements of the robot in the application	3	2	-	2
PC4. select the EOAT on the basis of its capability of handling maximum load	3	3	-	1
PC5. identify the zoning area and stroke area of robot by interpreting the process documents	2	3	-	1
PC6. determine the application controllers and external I/O devices required as per the project specifications and requirements	2	3	-	1
<i>Layout marking of robotic cell and tool assembly</i>	12	12	-	6
PC7. determine area required for system implementation and availability of power, pneumatic and coolant supply	4	4	-	2
PC8. plan material space, trolleys, supply of material to line side & material handling equipment	4	4	-	2
PC9. find the position of equipment and finalize the robot positions according to it	4	4	-	2
<i>Positioning of components in the cell</i>	12	12	-	6
PC10. identify the mounting and Tool Center Point (TCP) of equipment	4	4	-	2
PC11. finalize the required work tables of fixtures, orientation of loading and unloading and material flow in the robotic cell	4	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. mount and place all the components of robotic cell like robot, tip dressers, jigs/fixture/grippers, docking units, sensor and cable trays etc. as per the design document	4	4	-	2
NOS Total	40	40	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8317
NOS Name	Selection and setup of end-effector and robot
Sector	Automotive
Sub-Sector	Research & Development
Occupation	Automotive Product Development
NSQF Level	6
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Qualification Pack

ASC/N8318: Installation, commissioning and integration of robot system

Description

This NOS unit is about performing tasks related to robot installation, commissioning and teaching. It is also about creating obstacle free robotic paths and integrating robot controller for actual parameters.

Scope

The scope covers the following :

- Perform robot installation, commissioning and setup
- Carry out calibration and mastering of robot
- Perform robot teaching and testing

Elements and Performance Criteria

Perform robot installation, commissioning and setup

To be competent, the user/individual on the job must be able to:

- PC1.** install robot controller, licenses, tool, sensors and pneumatics into the system by following organisational procedures
- PC2.** integrate robot controller and robot as per SOP and design document
- PC3.** integrate safety fencing and controller panel as per SOP and design document
- PC4.** turn on the power of robot, do first operation, look for any warnings/errors in it and rectify the same as per organisational guidelines
- PC5.** check for sensors and external device connections with controller in case of any malfunction or no operation

Carry out calibration and mastering of robot

To be competent, the user/individual on the job must be able to:

- PC6.** execute mastering for all servos by following organisational procedures
- PC7.** define the global and local points (Home and Home 2) as per SOP
- PC8.** carry out tool configuration and data mapping in the system as per SOP
- PC9.** fix the TCP and mount the frame on tool as per design and project document
- PC10.** calibrate base of tool and record the readings for future reference

Perform robot teaching and testing

To be competent, the user/individual on the job must be able to:

- PC11.** insert instructions in the robotic system to execute teaching process
- PC12.** program the robot as per the path required for using point to point control system with necessary instructions
- PC13.** perform dry-run of the robot to check its functioning
- PC14.** perform all necessary tests and procedures required as per industry standards
- PC15.** override testing and motion types
- PC16.** create collision free path of the robot

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing and maintaining networks
- KU2.** software and system configuration
- KU3.** robot anatomy and operating system
- KU4.** EOAT anatomy and pneumatic systems
- KU5.** calibration and mastering processes
- KU6.** accuracy, speed and motion of robot
- KU7.** mechanism of linear and circular motion types
- KU8.** motion and time taken for diff activity

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform robot installation, commissioning and setup</i>	12	12	-	6
PC1. install robot controller, licenses, tool, sensors and pneumatics into the system by following organisational procedures	2	3	-	1
PC2. integrate robot controller and robot as per SOP and design document	3	2	-	2
PC3. integrate safety fencing and controller panel as per SOP and design document	3	2	-	1
PC4. turn on the power of robot, do first operation, look for any warnings/errors in it and rectify the same as per organisational guidelines	2	3	-	1
PC5. check for sensors and external device connections with controller in case of any malfunction or no operation	2	2	-	1
<i>Carry out calibration and mastering of robot</i>	14	14	-	6
PC6. execute mastering for all servos by following organisational procedures	3	3	-	2
PC7. define the global and local points (Home and Home 2) as per SOP	3	3	-	1
PC8. carry out tool configuration and data mapping in the system as per SOP	3	3	-	1
PC9. fix the TCP and mount the frame on tool as per design and project document	3	3	-	1
PC10. calibrate base of tool and record the readings for future reference	2	2	-	1
<i>Perform robot teaching and testing</i>	14	14	-	8
PC11. insert instructions in the robotic system to execute teaching process	3	3	-	2
PC12. program the robot as per the path required for using point to point control system with necessary instructions	3	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. perform dry-run of the robot to check its functioning	2	2	-	1
PC14. perform all necessary tests and procedures required as per industry standards	2	2	-	1
PC15. override testing and motion types	2	2	-	1
PC16. create collision free path of the robot	2	2	-	1
NOS Total	40	40	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8318
NOS Name	Installation, commissioning and integration of robot system
Sector	Automotive
Sub-Sector	Research & Development
Occupation	Automotive Product Development
NSQF Level	6
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Qualification Pack

ASC/N8319: Robot/Cobot programming and application testing

Description

This NOS unit is about programming and application parameterization of the robot/cobot.

Scope

The scope covers the following :

- Robot programming and parameterization
- Carry out application testing and dry run
- Operate robot on different modes

Elements and Performance Criteria

Robot programming and parameterization

To be competent, the user/individual on the job must be able to:

- PC1.** insert the instructions and define sequence of multiple paths/operation of the robot
- PC2.** modify path to achieve cycle time
- PC3.** assign application parameters in the program
- PC4.** create logics and insert variables for logical programming of the robot

Carry out application testing and dry run

To be competent, the user/individual on the job must be able to:

- PC5.** define parameters of robot application (welding/material handling)
- PC6.** connect application controllers with robot controller as per the layout diagram and robot manual
- PC7.** estimate the process path and cycle time as per production requirements
- PC8.** perform dry run of the robot on the job to check it functioning
- PC9.** fine tune the robot program with required cycle time and finalize the program

Operate robot on different modes

To be competent, the user/individual on the job must be able to:

- PC10.** check for safety door and interlocking systems for proper functioning
- PC11.** check safety fencing for proper functioning by applying T2 or AUT mode
- PC12.** perform dry run of robot on different operating modes
- PC13.** execute each operation on real job with all systems active
- PC14.** document the results in required formats by following organizational procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing and maintaining networks

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- KU2.** software and system configuration
- KU3.** robot anatomy and operating system
- KU4.** installation process includes robot mounting, wiring standards, routing, safety peripherals and tool integration
- KU5.** robot integration with automation elements like electro pneumatics and hydraulics, electrical components like circuit breakers, push buttons, sensors, relay, contactor, indicators, buzzer, motor conveyor, PLC, VFD, HMI
- KU6.** robot mastering, types and different conditions to do mastering
- KU7.** teach pendant controls and displays
- KU8.** robot programming methods, instructions using teach pendant and simulation tools
- KU9.** industry standards like safety device and its rating, wire and cable size capacity, connector types

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Robot programming and parameterization</i>	12	12	-	6
PC1. insert the instructions and define sequence of multiple paths/operation of the robot	3	3	-	2
PC2. modify path to achieve cycle time	3	3	-	1
PC3. assign application parameters in the program	3	3	-	1
PC4. create logics and insert variables for logical programming of the robot	3	3	-	2
<i>Carry out application testing and dry run</i>	14	14	-	8
PC5. define parameters of robot application (welding/material handling)	4	3	-	2
PC6. connect application controllers with robot controller as per the layout diagram and robot manual	3	3	-	1
PC7. estimate the process path and cycle time as per production requirements	3	3	-	2
PC8. perform dry run of the robot on the job to check it functioning	2	3	-	1
PC9. fine tune the robot program with required cycle time and finalize the program	2	2	-	2
<i>Operate robot on different modes</i>	14	14	-	6
PC10. check for safety door and interlocking systems for proper functioning	2	3	-	1
PC11. check safety fencing for proper functioning by applying T2 or AUT mode	3	3	-	2
PC12. perform dry run of robot on different operating modes	3	2	-	1
PC13. execute each operation on real job with all systems active	3	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. document the results in required formats by following organizational procedures	3	3	-	1
NOS Total	40	40	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8319
NOS Name	Robot/Cobot programming and application testing
Sector	Automotive
Sub-Sector	Research & Development
Occupation	Automotive Product Development
NSQF Level	6
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

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(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9810.Manage work and resources (Manufacturing)	50	30	-	20	100	15
ASC/N9812.Interact effectively with team, customers and others	50	30	-	20	100	10
ASC/N8316.Identify product feasibility and setup requirements	40	40	0	20	100	20
ASC/N8317.Selection and setup of end-effector and robot	40	40	0	20	100	20
ASC/N8318.Installation, commissioning and integration of robot system	40	40	0	20	100	15
ASC/N8319.Robot/Cobot programming and application testing	40	40	0	20	100	20
Total	260	220	-	120	600	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.